BTS(C) – VI – 09 – 009 – V

B. Tech Degree VI Semester Examination, April 2009

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

PART - A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) What are the three modes of Heat transfer? Explain their potential for occurrence.

(b) Define thermal conductivity and explain its significance in heat transfer.

(c) What are the differences between natural and forced convection?

(d) What is physical significance of Prandtl numbers?

(e) State and explain Planck's law of radiation.

(f) Explain the term 'radiation shape factor'.

(g) Compare parallel flow and counter flow heat exchanger.

(h) What are the limitations of LMTD Method? How is effectiveness – NTU method superior to LMTD method?

PART - B

(All questions carry ***EQUAL*** marks)

(4 x15 = 60)

II A wall is constructed of several layers. The first layer consists of brick (K= 0.66 W/m-K), 25cm thick, the second layer 2.5cm thick mortar (K = 0.7 W/m-K) , the third layer 10 cm thick limestone (K = 0.66 W/m-K) and the outer layer of 1.25cm thick plaster (K = 0.7 W/m.K). The heat transfer coefficients on interior and exterior of the wall are 5.8 $W/m^2 - K$ and 11.6 $W/m^2 - K$ respectively

Find : (i) Overall heat transfer coefficient.

(ii) Rate of heat transfer per unit area if the interior of the room is at 26^0 C while the outer air is at – 7^0C.

(iii) Temperature at the junctions.

OR

III. If a thin and long fin, insulated at its tip is used, show that the heat transfer from the fin is given by $Q_{fin} = \sqrt{hpk\,Ac}\left(T_o - T_\infty\right)\tan h\left(ML\right)$.

IV. Air at 10^0C and at a pressure of 100kPa is flowing over a flat plate at a velocity of 3 m/s. If the plate is 30cm wide and at a temperature of 60^0C, calculate the flowing at a distance of x = 0.3 m from the leading edge.

(i) Thickness of velocity and thermal boundary layers.

(ii) Local and average friction coefficients.

(iii) Local shear stress.

(iv) Total drag force.

(v) Local and average heat transfer coefficients.

(vi) Total heat transfer from the plate,

OR

(Turn Over)

V. A hot plate 1 m x 0.5 m at 130^0 C kept vertically in still air at 20^0 C. Find -

(i) Heat transfer coefficient
(ii) Initial rate of cooling the plate in ^{0}C/min,
(iii) Time required for cooling the plate from 180^0C to 80^0C if the heat transfer is due to convection only.

Mass of the plate is 20Kg and $C_p = 400$ J/Kg-K. Assume 0.5m side vertical and that the heat transfer coefficient remains constant and convection takes place from both sides of the plate.

VI. Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500^0 C.

(i) Monochromatic emissive power at 1.2 μ m length
(ii) Wavelength at which the emission is maximum.
(iii) Maximum emissive power.
(iv) Total emissive power
(v) Total emissive power of the furnace if it is assumed as a real surface with emissivity equal to 0.9.

OR

VII. The large parallel planes with emissivities 0.3 and 0.8 exchange heat. Find the percentage reduction when a polished aluminum shield of emissivity 0.04 is placed between them. Use the method of electrical analogy.

VIII. Derive an expression for log mean temperature difference (LMTD) for parallel flow heat exchanger. How this expression can be modified for counter flow heat exchanger.

OR

IX. A chemical having specific heat of 3.3kJ/Kg-K at a rate of 20,000 kg/h enters a parallel flow heat exchanger at 120^0C. The flow rate of cooling water is 50,000 kg/hr with an inlet temperature of 20^0C. The heat transfer area is $10m^2$ and overall heat transfer coefficient is 1050 W/m^2 – K. Find

(i) The effectiveness of the heat exchanger
(ii) Outlet temperatures of water and chemical.
Take C_p of water as 4.186kJ/kg-K.

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